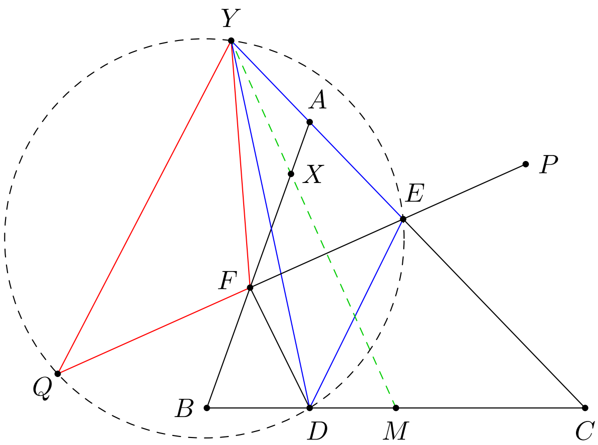
HK IMO Training 2021-2022
Problem Set 26 Suggested Solutions

Problem 101 (Czech and Slovak MO Round III Category A 2016 Problem 3)**.** There are some clubs in a school. Every student can join any number of clubs they wish. It is known that every pair of clubs has at least one member in common. Prove that it is possible to put the students into two societies, such that at most one student belongs to both societies, and each society consists of students from every club.

Solution. Consider a club C with the smallest number of members. Suppose we put every student in C into the first society, and put all other students into the second society. By the condition, every club must contain a member which belongs to C. So the first society consists of students from every club. The only clubs which have no members belonging to the second society are those which do not contain members other than those in C. By the minimality of the number of members of C, we see that the members of those clubs must be identical to the members of C. Thus, if we put an arbitrary member of C in the second society as well, we are done. $\square$

Problem 102 (Slovenia TST 2019 Day 1 Problem 5)**.** In $\triangle ABC$, let D, E, F be the feet of altitudes from A, B, C respectively. Points P and Q are chosen on the line EF such that Q, F, E, P lie in this order, $EP = DF$ and $FQ = DE$. Point X is chosen on the line AB such that $XD = XP$. Point Y is chosen on the line AC such that $YD = YQ$. Prove that the line XY bisects BC.



Solution. Since $\angle AEF = \angle ABC = \angle CED$, the line EY is the external angle bisector of $\angle QED$. As $YD = YQ$, the points D, E, Y, Q are concyclic. This implies $\angle EQY = \angle EDY$. Together with $DE = QF$ and $YD = YQ$, we have $\triangle YQF \cong \triangle YDE$. Thus, $YF = YE$. Similarly, $XF = XE$. Also, it is well-known that $MF = ME$ where M is the midpoint of BC (as B, C, E, F lie on a circle with centre M). So Y, X, M lie on the perpendicular bisector of FE, which means XY bisects BC. $\square$

Problem 103 (Saudi Arabia TST for GMO 2014 Problem 8)**.** Let $x_1 \geqslant x_2 \geqslant \cdots \geqslant x_n$ be positive real numbers. Prove that

$$x_1x_2(x_1 - x_2) + x_2x_3(x_2 - x_3) + \cdots + x_{n-1}x_n(x_{n-1} - x_n) \geqslant x_1x_n(x_1 - x_n).$$

Solution. We prove this by induction on $n \geqslant 2$. For $n = 2$, clearly both sides are the same. Assume this inequality is true for some n. For the case $n + 1$, we have

$$x_1x_2(x_1 - x_2) + x_2x_3(x_2 - x_3) + \cdots + x_{n-1}x_n(x_{n-1} - x_n) + x_nx_{n+1}(x_n - x_{n+1}) \geqslant x_1x_n(x_1 - x_n) + x_nx_{n+1}(x_n - x_{n+1})$$

by the inductive hypothesis. Now, we note that

$$x_1x_n(x_1 - x_n) + x_nx_{n+1}(x_n - x_{n+1}) - x_1x_{n+1}(x_1 - x_{n+1}) = (x_1 - x_n)(x_n - x_{n+1})(x_1 - x_{n+1}) \geqslant 0.$$

Combining these, the case $n + 1$ is established. The proof is complete by induction. $\square$

Problem 104 (Saudi Arabia TST for GMO 2015 Problem 8)**.** For any positive integer n and any nonnegative integer $k \leqslant n$, let r_k be the remainder when $\binom{n}{k}$ is divided by 3. Find all positive integers n such that $r_0 + r_1 + \cdots + r_n \geqslant n$.

Solution. Let $k = \overline{k_m k_{m-1} \cdots k_0}_{(3)}$ and $n = \overline{n_m n_{m-1} \cdots n_0}_{(3)}$ be the base 3 representations of k and n respectively (in the following, we drop the index whenever we use the base 3 representation). By Lucas' theorem, we have

$$r_k \equiv \binom{n}{k} \equiv \prod_{i=0}^{m} \binom{n_i}{k_i} \pmod 3.$$

If there exists $0 \leqslant i \leqslant m$ such that $k_i > n_i$, then $\binom{n_i}{k_i} = 0$ and so $r_k = 0$. Therefore, we may assume $k_i \leqslant n_i$ for all $0 \leqslant i \leqslant m$.

For each i, if $n_i = 0$, then $k_i = 0$. If $n_i = 1$, then $\binom{n_i}{k_i} = 1$ no matter $k_i = 0$ or 1. If $n_i = 2$, then $\binom{n_i}{k_i} = 1$ if $k_i \neq 1$ and $\binom{n_i}{k_i} = 2$ if $k_i = 1$. Therefore, if the number a of pairs $(n_i, k_i) = (2, 1)$ is odd, then $\binom{n}{k} \equiv 2^a \equiv 2 \pmod 3$, otherwise $\binom{n}{k} \equiv 1 \pmod 3$. It follows that

$$s(n) := r_0 + r_1 + \cdots + r_n = 2A(n) + B(n),$$

where $A(n)$ is the number of pairs (n, k) such that $0 \leqslant k_i \leqslant n_i$ for $0 \leqslant i \leqslant m$, and the number of pairs $(n_i, k_i) = (2, 1)$ is odd, while $B(n)$ is the number of pairs (n, k) such that $0 \leqslant k_i \leqslant n_i$ for $0 \leqslant i \leqslant m$, and the number of pairs $(n_i, k_i) = (2, 1)$ is even.

Denote by r, s, t the numbers of $0, 1, 2$ in $n_0, n_1, \ldots, n_m$ respectively. For those $n_i = 1$, there are 2 choices of k_i. Among the t copies of 2 in $n_0, n_1, \ldots, n_m$, we have to choose a of them such that the corresponding k_i is 1, where a is odd, and there are 2 choices of k_i for the other indices. Thus, we see that

$$A(n) = 2^s \sum_{\substack{a=0 \\ 2 \nmid a}}^{t} \binom{t}{a} 2^{t-a}.$$

Similarly, we have

$$B(n) = 2^s \sum_{\substack{b=0 \\ 2 \mid b}}^{t} \binom{t}{b} 2^{t-b}.$$

It follows that

$$\begin{aligned} s(n) &= 2^s \sum_{\substack{j=0 \\ 2 \mid j}}^{t} \left[\binom{t}{j} + \binom{t}{j+1} \right] 2^{t-j} = 2^s \sum_{\substack{j=0 \\ 2 \mid j}}^{t} \binom{t+1}{j+1} 2^{t-j} \\ &= 2^s \cdot \frac{(2+1)^{t+1} - (2-1)^{t+1}}{2} = 2^{s-1}(3^{t+1} - 1). \end{aligned}$$

- If $n_m = 2$, then

$$\begin{aligned} n &\geqslant \overline{2\underbrace{00\cdots0}_{r}\underbrace{11\cdots1}_{s}\underbrace{22\cdots2}_{t-1}} \\ &= 2 \cdot 3^{r+s+t-1} + \frac{3^s - 1}{2} \cdot 3^{t-1} + 2 \cdot \frac{3^{t-1} - 1}{2} \\ &= \frac{4 \cdot 3^{r+s} + 3^s + 1}{2} \cdot 3^{t-1} - 1. \end{aligned}$$

  Therefore,

$$n - s(n) \geqslant \frac{(4 \cdot 3^r + 1) \cdot 3^s - 9 \cdot 2^s + 1}{2} \cdot 3^{t-1} - 1 + 2^{s-1}.$$

  - If $r \geqslant 1$, then $n - s(n) \geqslant \frac{13 - 9 + 1}{2} \cdot \frac{1}{3} - 1 + \frac{1}{2} > 0$.
  - If $r = 0$ and $s \geqslant 2$, then $n - s(n) \geqslant \frac{45 \cdot 3^{s-2} - 36 \cdot 2^{s-2} + 1}{2} \cdot 3^{t-1} - 1 + 2 > 0$.
  - If $r = 0$ and $s = 1$, then $n \leqslant \overline{\underbrace{22\cdots2}_{t}1} = 3^{t+1} - 2 < 3^{t+1} - 1 = s(n)$.

- If $r = s = 0$, then $n = \overline{\underbrace{22\cdots 2}_{t}} = 3^t - 1 < \dfrac{3^{t+1} - 1}{2} = s(n)$.

- If $n_m = 1$, then

$$\begin{aligned} n &\geqslant \overline{1\underbrace{00\cdots 0}_{r}\underbrace{11\cdots 1}_{s-1}\underbrace{22\cdots 2}_{t}} \\ &= 3^{r+s+t-1} + \frac{3^{s-1} - 1}{2} \cdot 3^t + 2 \cdot \frac{3^t - 1}{2} \\ &= \frac{2 \cdot 3^{r+s-1} + 3^{s-1} + 1}{2} \cdot 3^t - 1. \end{aligned}$$

  Therefore,

$$n - s(n) \geqslant \frac{(2 \cdot 3^r + 1) \cdot 3^{s-1} - 6 \cdot 2^{s-1} + 1}{2} \cdot 3^t + 2^{s-1} - 1.$$

  - If $r \geqslant 1$, then $n - s(n) \geqslant \dfrac{7 \cdot 3^{s-1} - 6 \cdot 2^{s-1} + 1}{2} \cdot 3^t + 2^{s-1} - 1 \geqslant 1 + 1 - 1 > 0$.
  - If $r = 0$ and $s \geqslant 3$, then $n - s(n) \geqslant \dfrac{3(3^{s-1} - 2^s) + 1}{2} \cdot 3^t + 2^{s-1} - 1 \geqslant 2 + 4 - 1 > 0$.
  - If $r = 0$ and $s = 2$, then $n \leqslant \overline{1\underbrace{22\cdots 2}_{t}1} = 2(3^{t+1} - 1) = s(n)$.
  - If $r = 0$ and $s = 1$, then $n = \overline{1\underbrace{22\cdots 2}_{t}} = 3^{t+1} - 3^t - 1 < 3^{t+1} - 1 = s(n)$.

Summarizing all these, we see that $n = 2 \cdot 3^a - 3^b - 1$ for some $a \geqslant b \geqslant 1$, or $n = 3^a - 3^b - 1$ for some $a > b \geqslant 1$. $\square$